UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

4622757166

CHEMISTRY **5070/04**

Paper 4 Alternative to Practical **October/November 2009**

1 hour

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

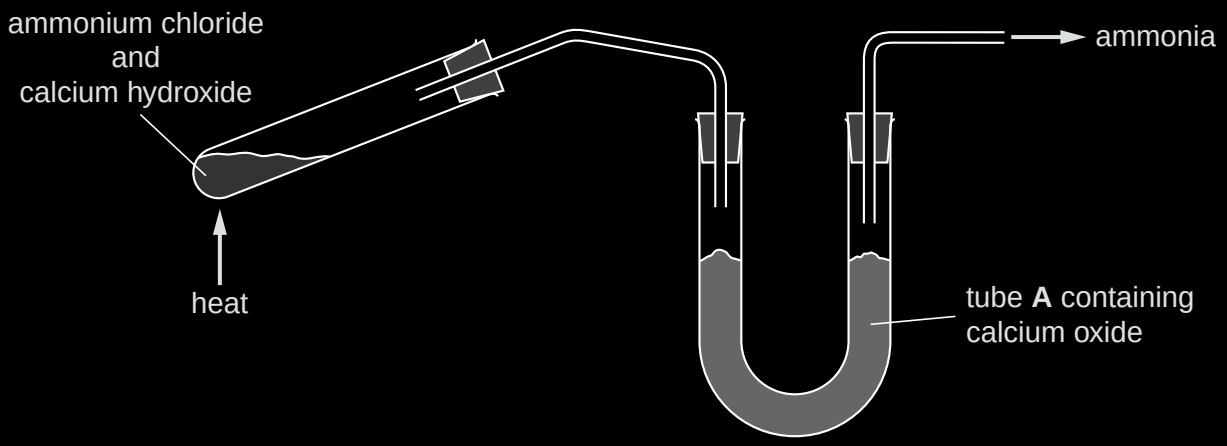
Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Write your answers in the spaces provided in the Question Paper.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use

This document consists of **19** printed pages and **1** blank page.

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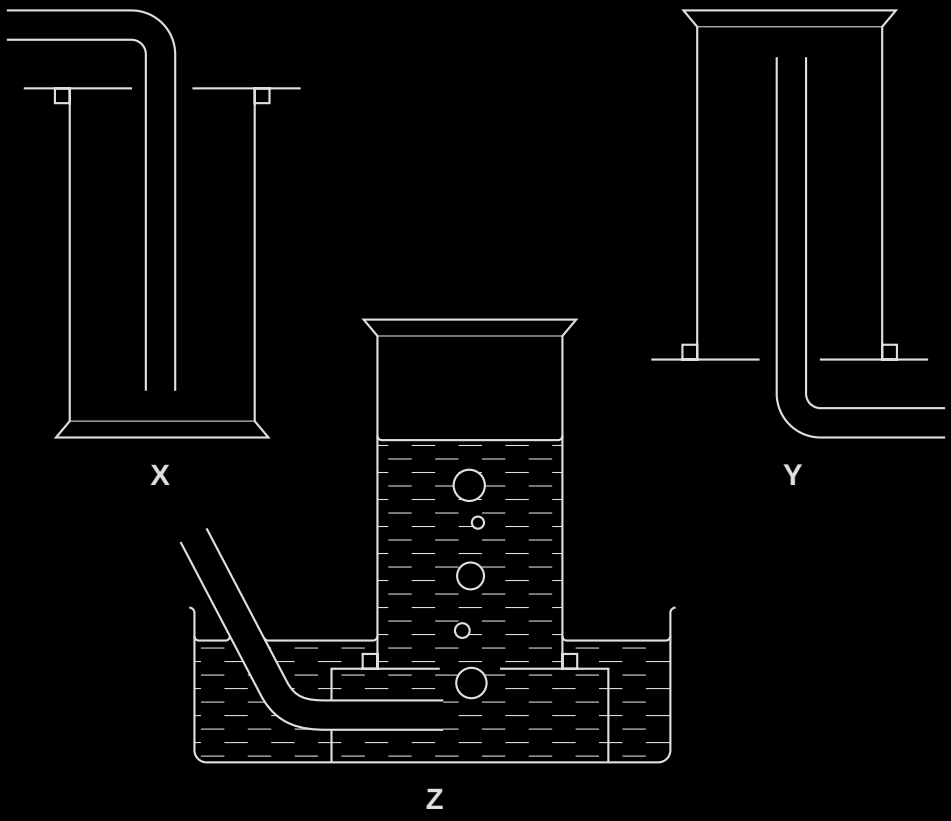


The equation for the reaction is

$$Ca(OH)_2 + 2NH_4Cl \rightarrow CaCl_2 + 2H_2O + 2NH_3$$

(a) By referring to the equation suggest why tube **A**, which contains calcium oxide, is included in the apparatus.

..... [1]

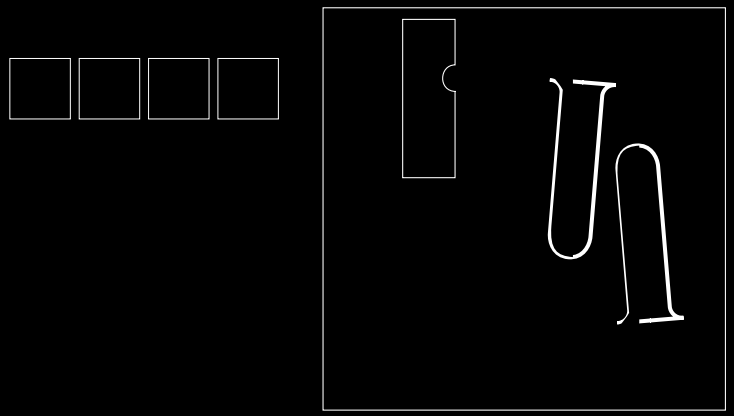


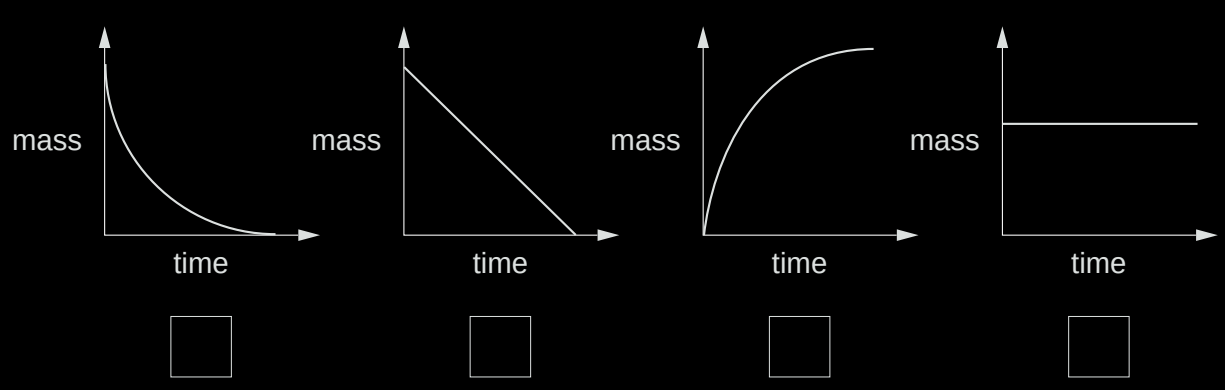
method of collection .....

explanation .....

..... [3]

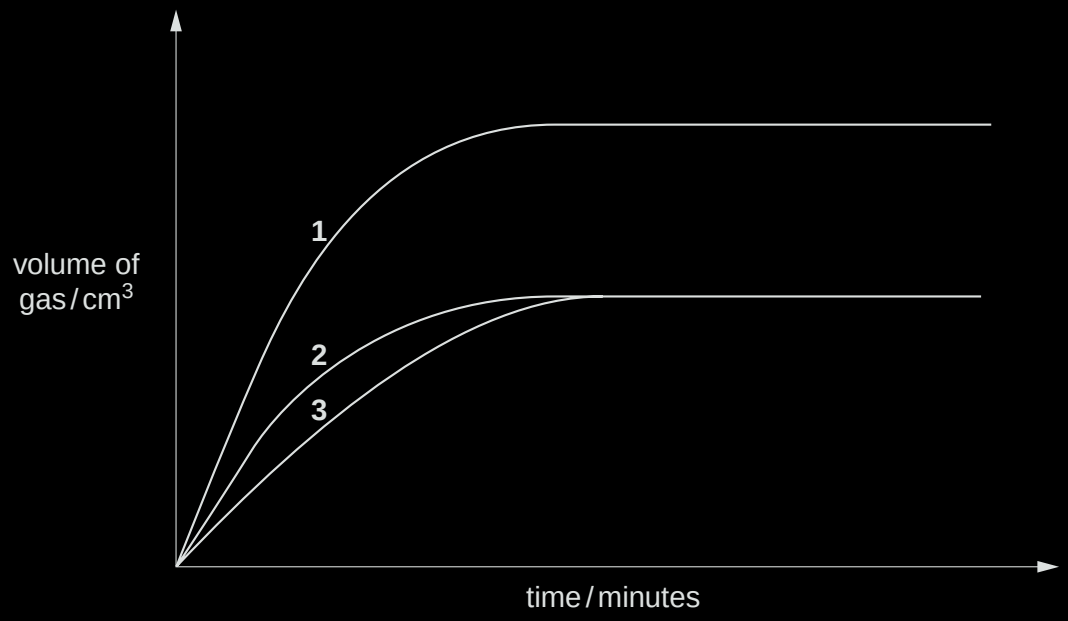






(a) **(b)** **(c)** **(d)**

[Total: 1]



Which statement is correct?

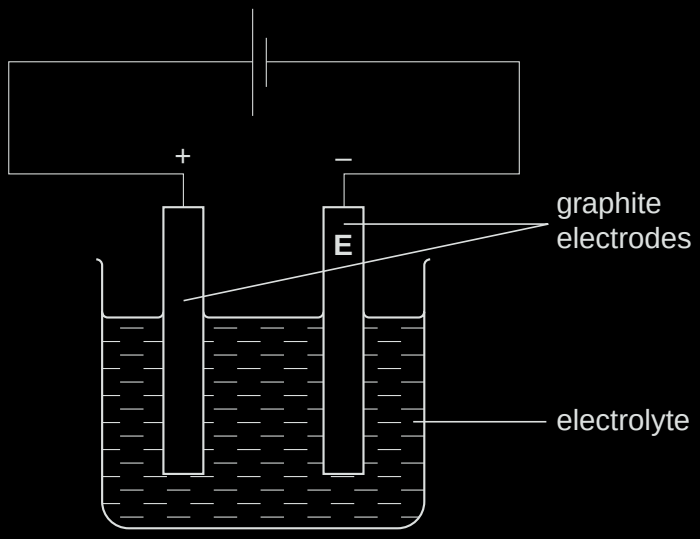
(a) In experiment **1** the number of moles of acid was less than in experiment **2**. ☐

(b) In experiment **3** the calcium carbonate was more finely powdered than in experiment **1**. ☐

(c) In experiments **2** and **3** the number of moles of acid was the same. ☐

(d) In experiment **3** the concentration of the acid was higher than in experiment **2**. ☐

[Total: 1]

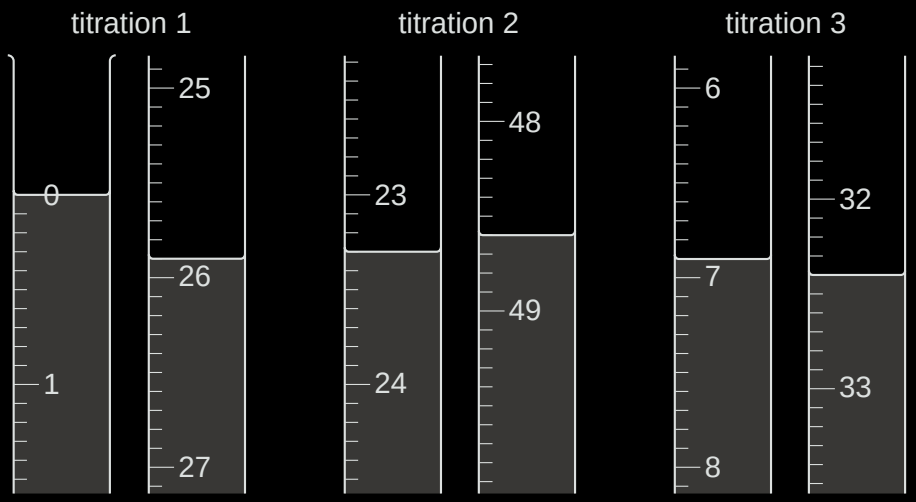


In experiment **1** the electrolyte was molten sodium chloride and in experiment **2** it was concentrated aqueous sodium chloride.

What were the products at the electrode labelled **E**?

	experiment **1** molten sodium chloride	experiment **2** concentrated aqueous sodium chloride	
(a)	chlorine	oxygen	☐
(b)	sodium	hydrogen	☐
(c)	chlorine	hydrogen	☐
(d)	sodium	oxygen	☐
(e)	sodium	chlorine	☐

[Total: 1]



(d) Use the diagrams to complete the following table.

titration number	1	2	3
final burette reading / cm^3			
initial burette reading / cm^3			
volume of 0.100 mol / dm^3 sodium hydroxide / cm^3			
best titration results (✓)			

Summary:

Tick (✓) the best titration results.

Using these results, the average volume of 0.100 mol / dm^3

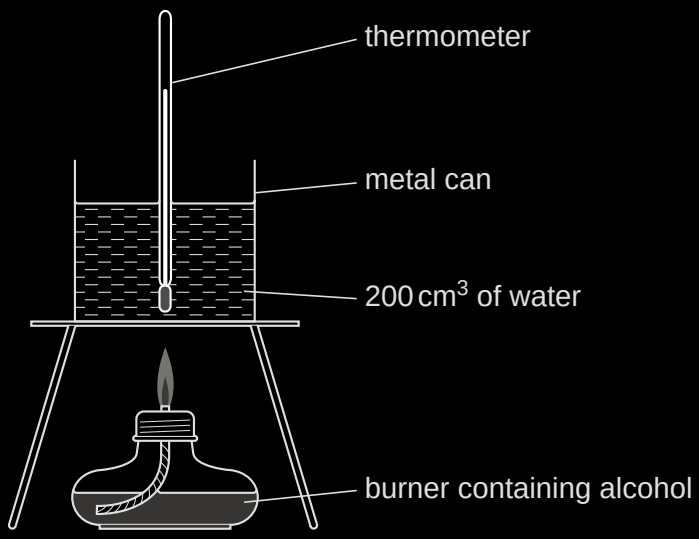
sodium hydroxide was ..... cm^3. [4]

(e) Calculate the number of moles of sodium hydroxide in the average volume of 0.100 mol / dm^3 sodium hydroxide in **(d)**.

..... moles [1]



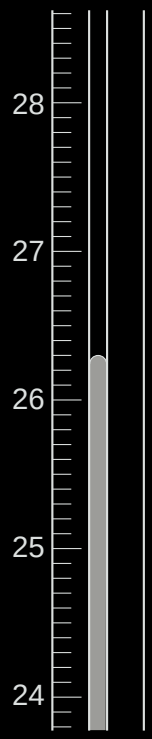




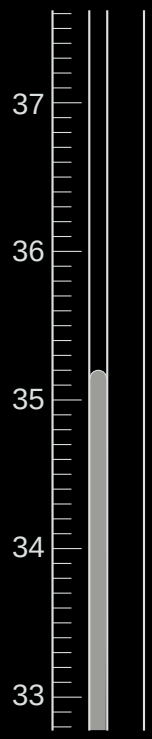
Some propanol was put into the burner which was then weighed.
The temperature of the water was noted.
The burner was lit and allowed to burn for several minutes.
The flame was extinguished and the final temperature of the water was noted. The burner was reweighed.

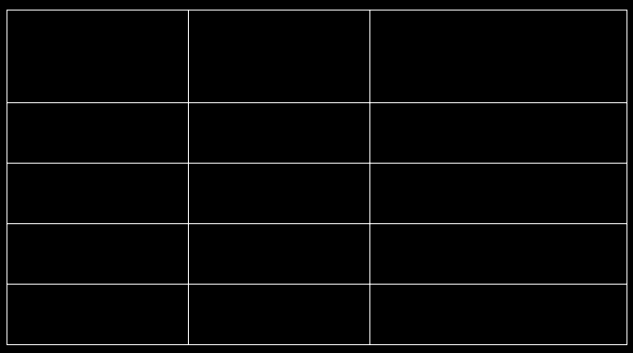
The diagrams below show parts of the thermometer stem for each of the temperature readings.

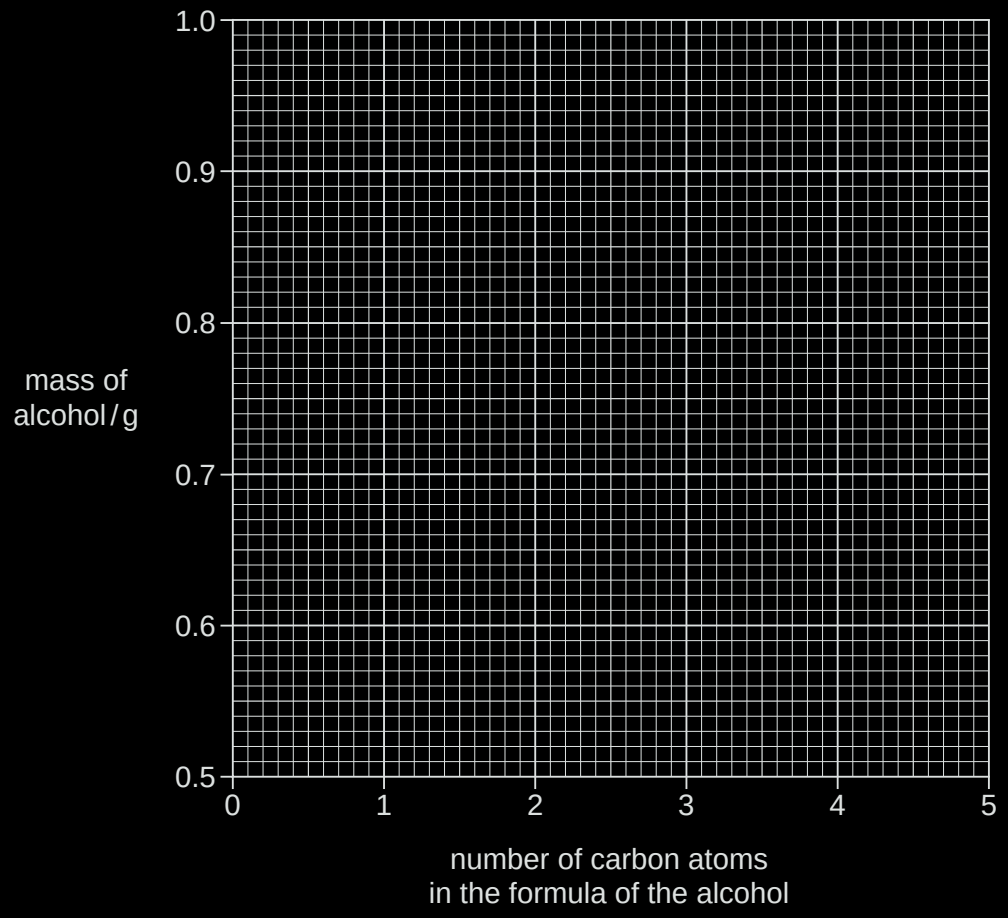
initial temperature



final temperature







[2]

(f) Predict the mass of butanol, C_4H_9OH, which, on combustion, would raise the temperature of the water by 15 °C.

..... g [1]

[Total: 12]